

PORTFOLIO 2026

Tan Zhi Xiang

Mechanical and electronic design engineer

Over 2 years building interactive exhibits and installations for science centres, museums and public events in Singapore, plus robots, web apps and AI tools of my own. 22 projects follow.

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Projects

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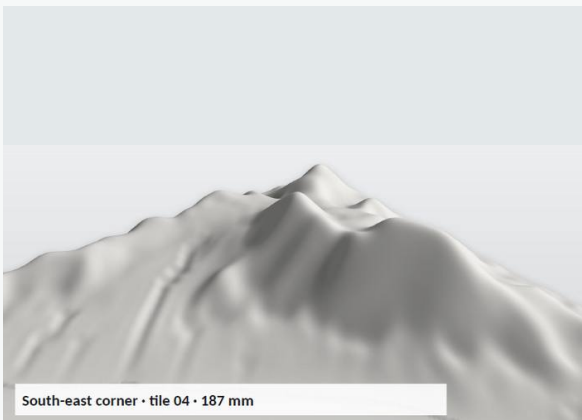
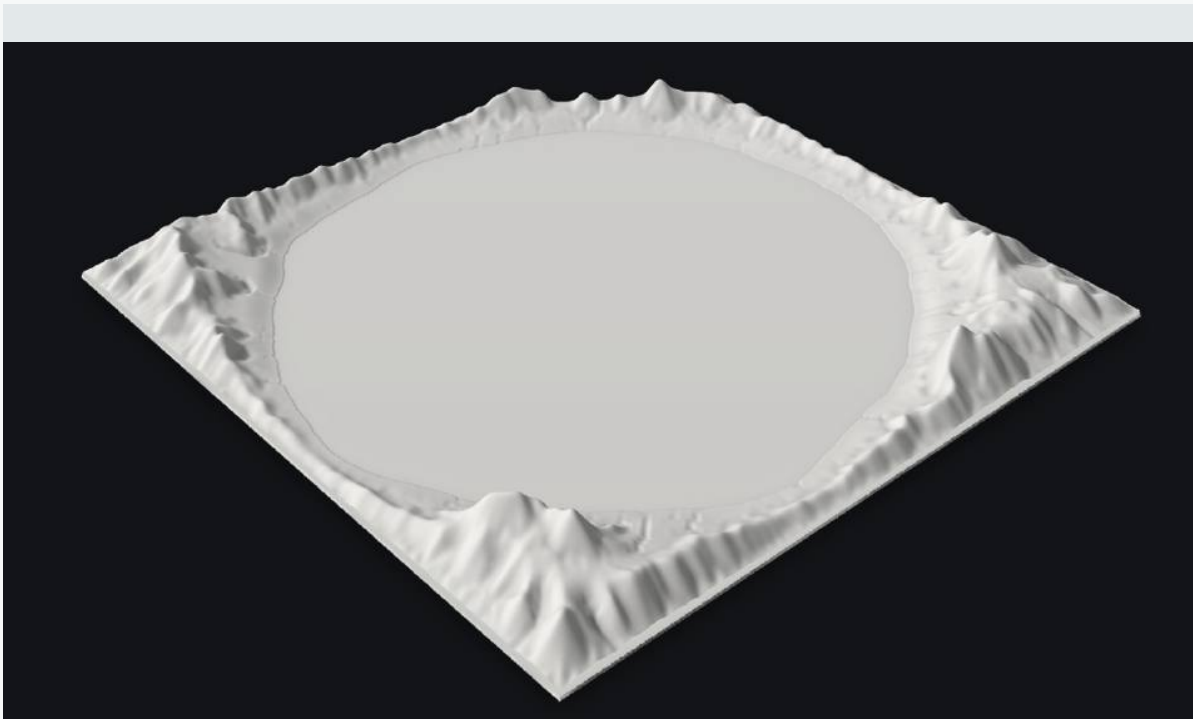
OTHER | WORK

Terrain Diorama

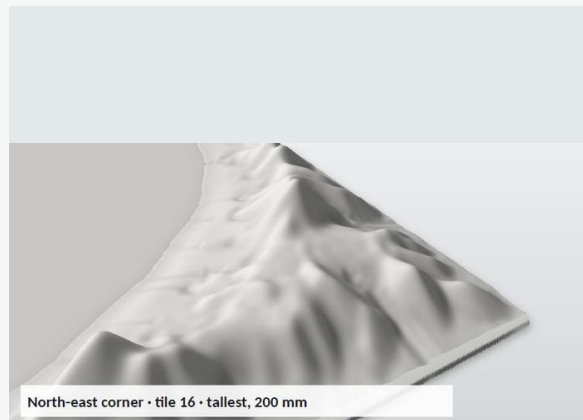
Setting: Romance of the Three Kingdoms theme

The physical terrain for a 3 m x 3 m projection diorama: four organic corner terrain pieces around a flat central circle that takes the projection.

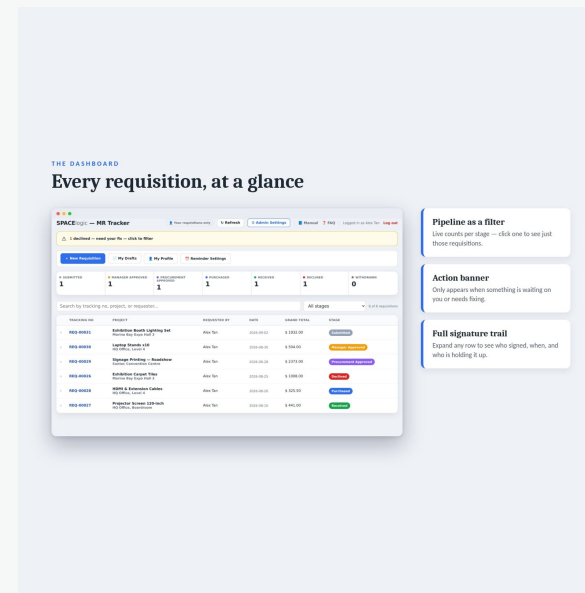
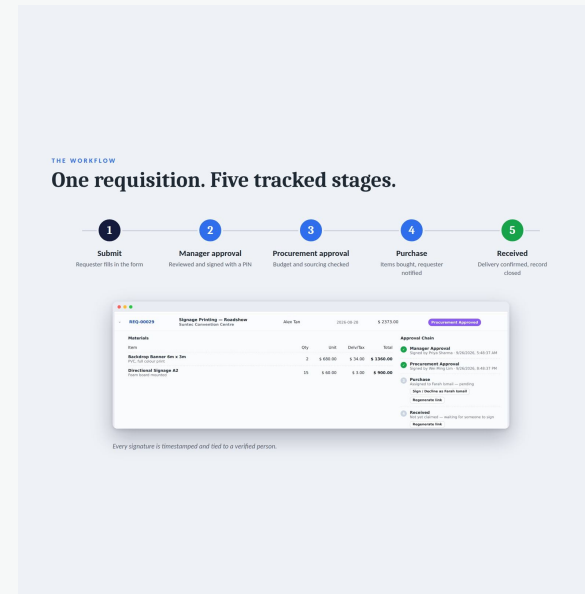
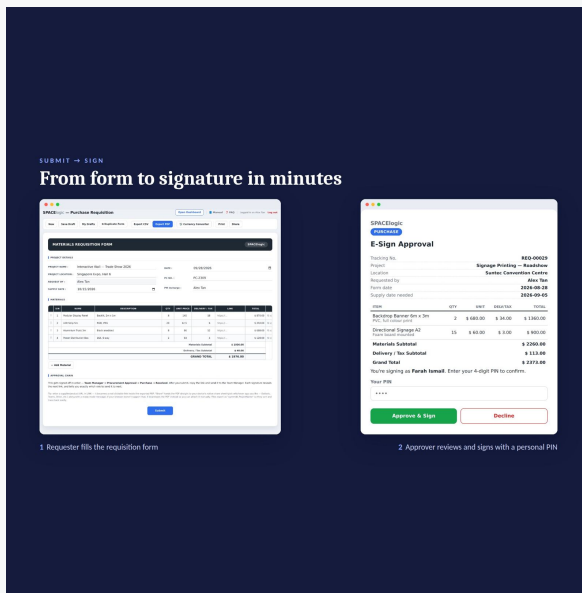
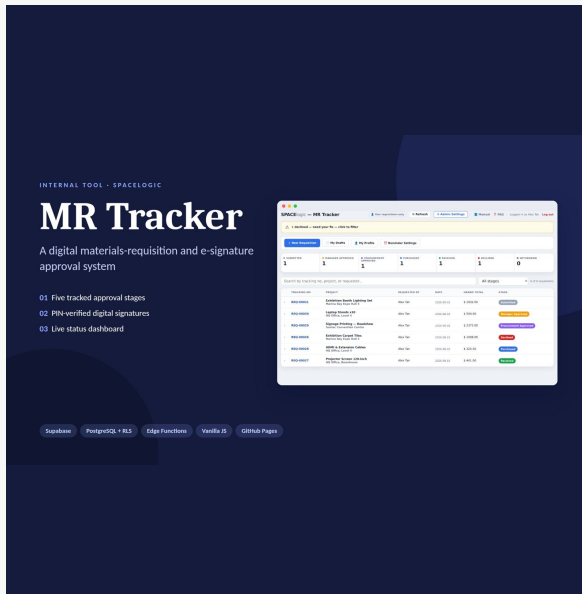
I used AI-assisted 3D modelling and Fusion 360 to shape the mountains, ridges and rivers, refined the model for fabrication, and produced the sections on a large-format BigRep 3D printer.



South-east corner • tile 04 • 187 mm



North-east corner • tile 16 • tallest, 200 mm



SOFTWARE | WORK

MR Tracker

Setting: SPACElogic internal tool

A web-based materials requisition and e-signature approval system for SPACElogic. Every request follows one fixed chain (Submit, Manager Approval, Procurement Approval, Purchase, Received), and each approval is a timestamped signature tied to a verified person.

When several people have to approve a purchase in order, the hard parts are visibility (where is it stuck?) and accountability (who approved what, and when?). I wanted one place where a requester can see exactly where a request stands, and where every signature leaves a permanent trail.

Built with: HTML, CSS, vanilla JavaScript; Supabase (PostgreSQL, row-level security, PL/pgSQL, Edge Functions); Resend; GitHub Pages; jsPDF

What I built

- **Requisition form** with drafts, running totals, drag-to-reorder rows, duplicate row and duplicate form, an in-form currency converter, and PDF export (including Chinese text).
- **Sequential e-signature chain** where approvers sign through a link with a personal PIN, and the order is enforced on the server so steps cannot be skipped.
- **Live dashboard** with stage counts that double as filters, a banner that only appears when something needs your attention, and an expandable signature trail for every requisition.
- **Decline, revise and resubmit:** declined requests reopen in the full form, and resubmitting restarts the chain so everyone re-confirms the revised request.
- **Withdraw, reminders and notifications:** requesters can withdraw before purchase, set their own overdue thresholds per stage, and get emailed on decline, purchase and receipt.
- **Admin tools** to manage accounts and roles, and to override or reassign a stuck stage.
- **Built-in user manual and FAQ** for onboarding.

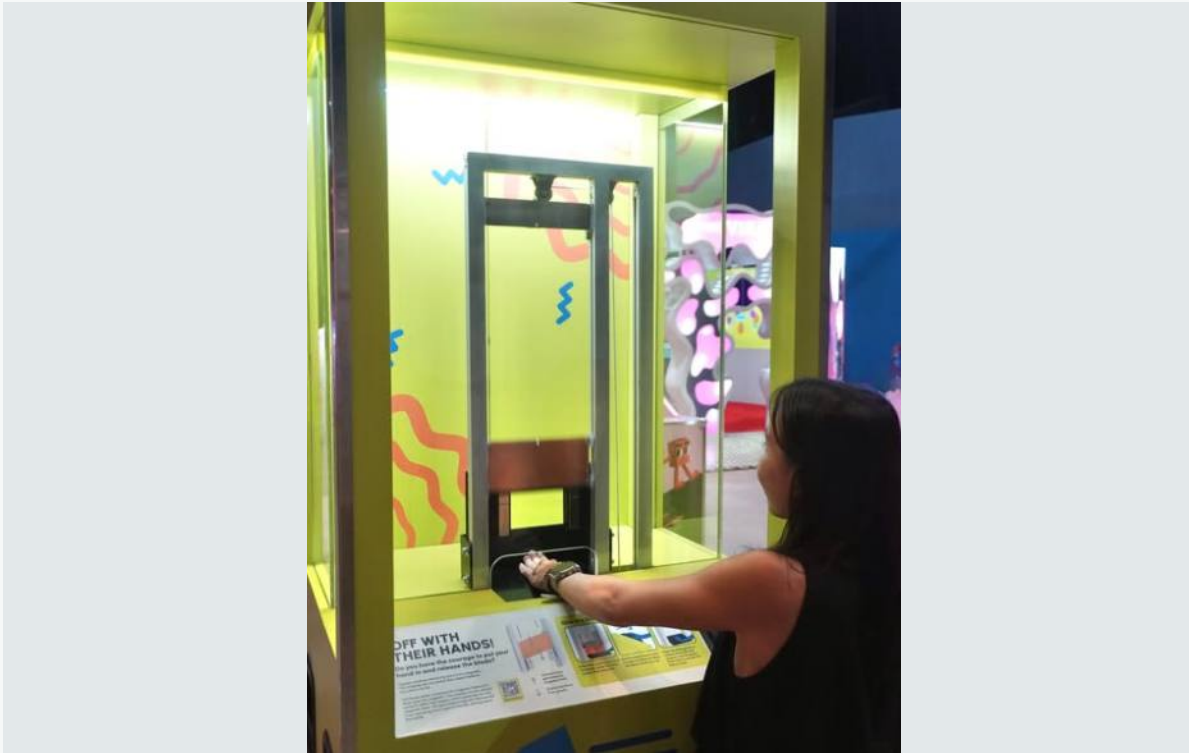
Design decisions

- **Security lives in the database, not the page.** Tables are closed to direct access and everything goes through functions that check identity and role, so hiding a button is never the only protection.
- **PINs are hashed with bcrypt**, and signing locks after repeated wrong attempts.

- **Accounts are tied to a verified company email**, one account per address.
- **Low running cost by design:** it runs on the free tiers of GitHub Pages, Supabase and Resend, with a clear upgrade path if usage grows.

Problems I had to solve

- **Chinese text disappeared from exported PDFs.** The PDF library's built-in fonts only cover Latin characters, so I detected CJK text and rendered just those strings as high-resolution images, keeping everything else as normal vector text.
- **Signature boxes vanished on longer forms.** With about 10 line items the totals and sign-off sections were drawn below the page edge. I traced the layout maths and added page-break handling.
- **Drag-and-drop reordering was unreliable** with the browser's native drag events, so I rebuilt it on plain mouse events.
- **Notification emails were delivered but never arrived.** A brand-new sending domain was being silently held by the company mail filter. I set up a verified domain (SPF and DKIM) and documented what IT needs to allowlist.



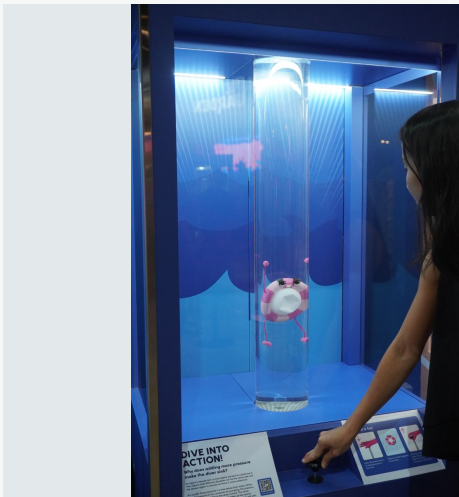
MECHANICAL | WORK

SCS Forced to Go

Setting: Science Centre Singapore

A travelling exhibition that brings science to life through hands-on experiments on forces, magnetism, buoyancy, pressure and electrostatics.

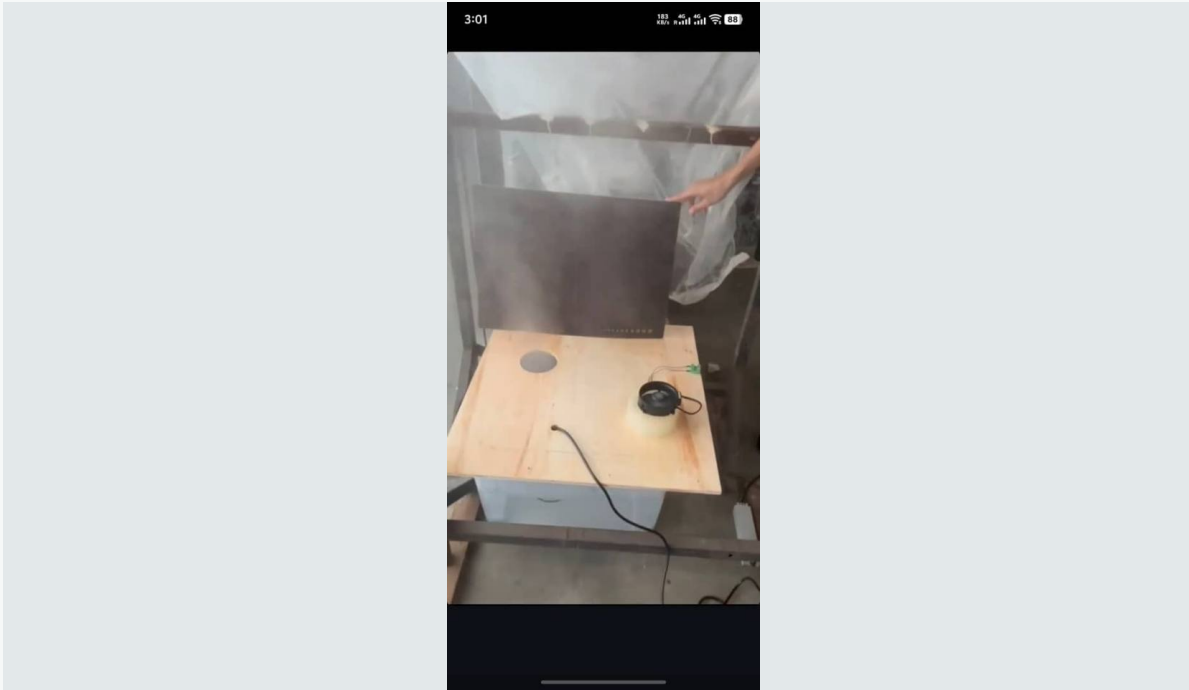
Exhibits I worked on include buoyancy changing with pressure, a falling copper plate slowed by a magnetic field, and a slinky bouncing down a moving treadmill.



Buoyancy changing with pressure



Slinky bouncing down a moving treadmill



Prototype for proving the concept

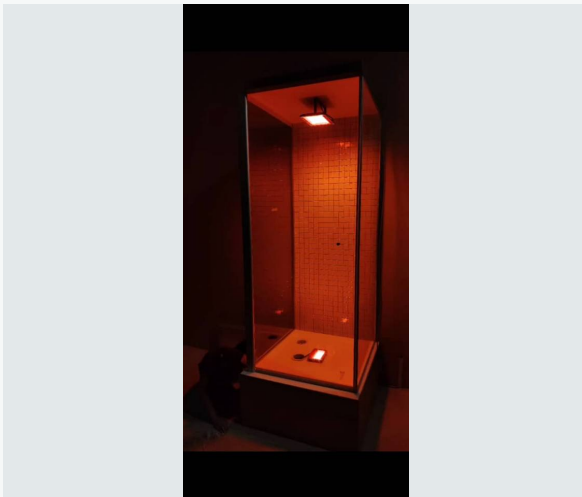
MECHANICAL | WORK

SAM Misting Machine

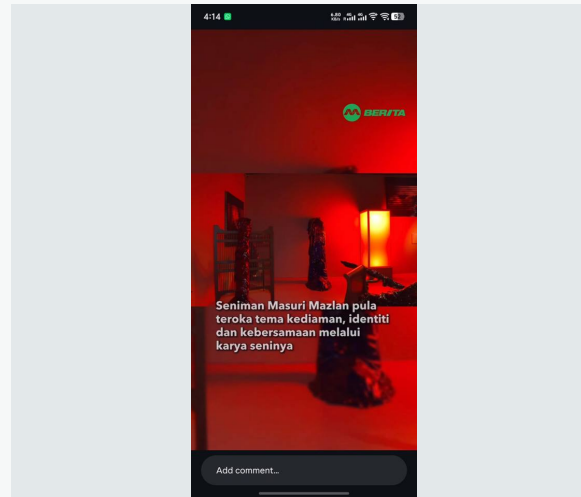
Setting: ArtScience Museum, artist Masuri Mazlan

A misting shower unit for an artwork by Masuri Mazlan. The water circulates on its own to make the mist, and lighting carries the artist's intent.

I designed it from a proof-of-concept prototype through to the final piece.



Final look before power on



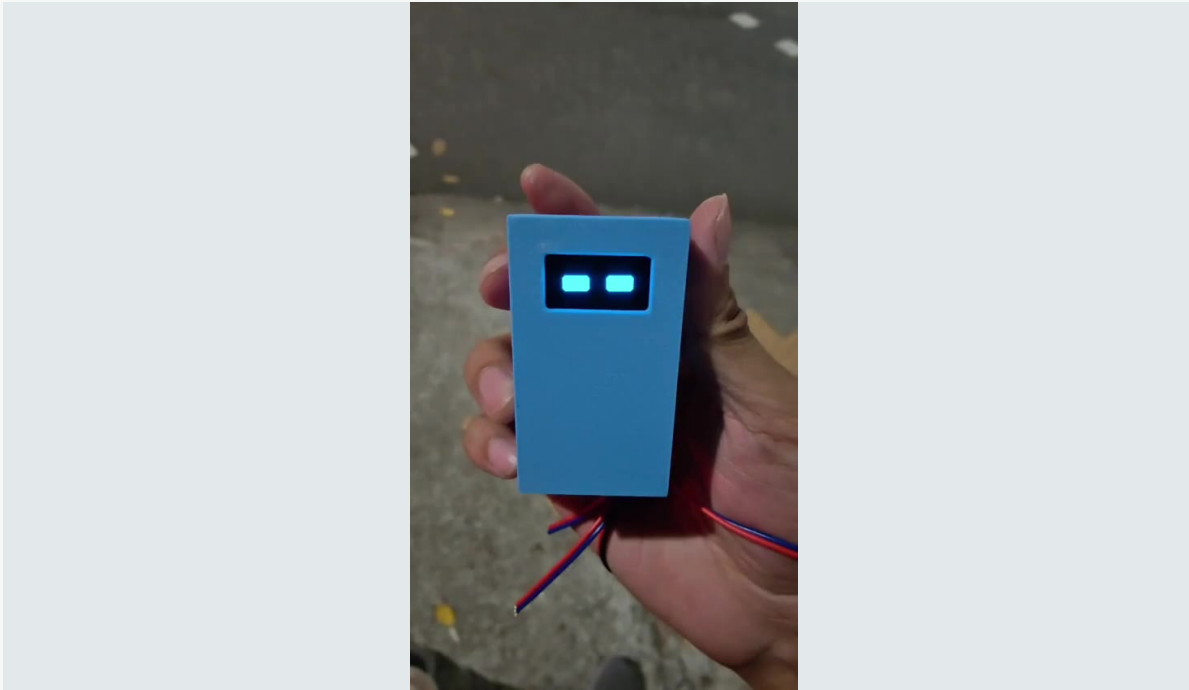
Final look with the misting effect

Robot Pet (Cubot)

Setting: ESP32, OLED face, Bluetooth app

A palm-sized desk robot that responds to touch and head rubbing with vibration and expressive facial animations.

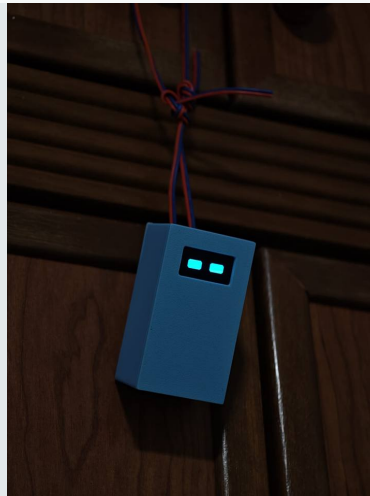
It runs on an ESP32 with an OLED face, a touch sensor, a vibration motor and an MPU-6050 motion sensor, and pairs with a phone app over Bluetooth to sync the time and change settings. There is a mini cube version and a portable version.



Product showcase



Mini cube version



Portable version

Video of this project is on the online portfolio.

Chalkline

Setting: React, Supabase, GitHub Pages

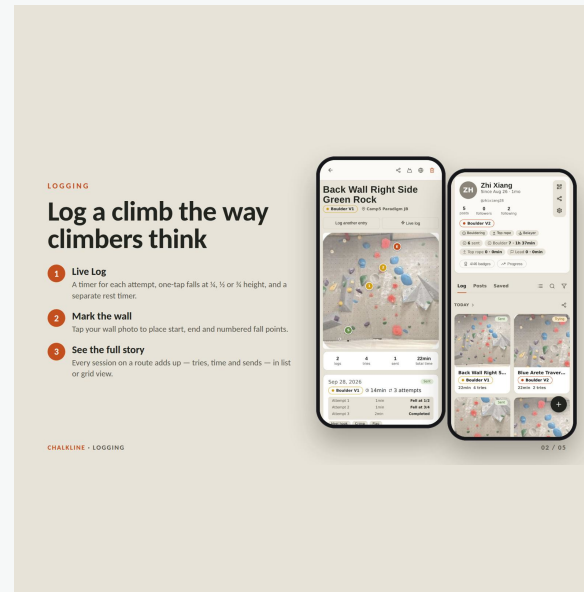
A climbing log and community app. Log every climb, track your progress and share it with your crew.

I built it as a real app for real use, with email sign-up, private logs kept private by the database, and account deletion that removes everything.

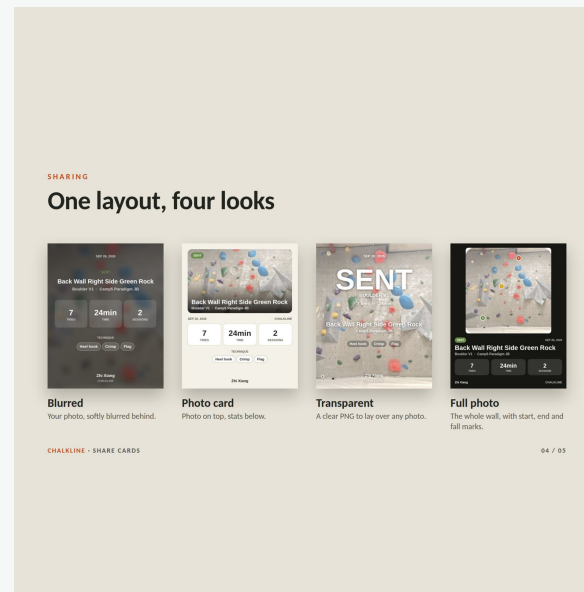
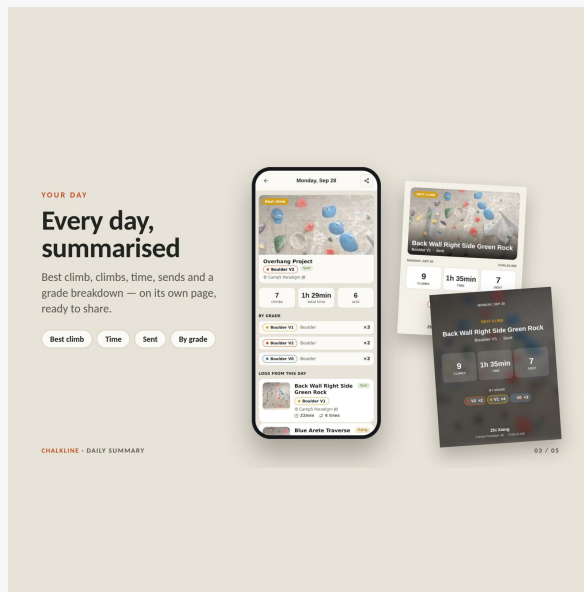
Built with: React; Supabase (Postgres, Auth, Edge Functions); GitHub Pages



Chalkline: climbing log and community app



Logging a climb with a live attempt timer and photo markers



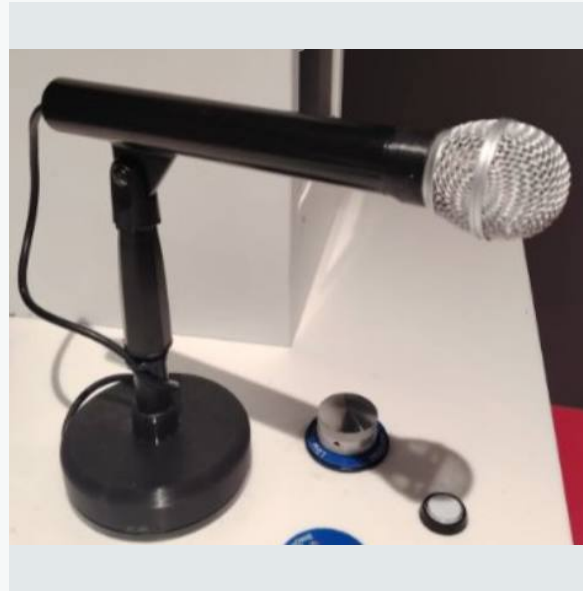
Share cards: one layout, four looks

Features

- **Live Log:** an attempt timer, one-tap falls at 1/4, 1/2 or 3/4 height, and a separate rest timer.
- **Photo markers:** tap your wall photo to place start, end and numbered fall points.
- **Full story:** every session on a route adds up (tries, time and sends), in list or grid view.
- **Daily summary:** best climb, climbs, time, sends and a grade breakdown on its own page, ready to share.
- **Share cards:** log, day, badge, progress and name tag cards generated in the browser, in four looks: blurred, photo card, transparent PNG and full photo.
- **Community:** feed, gear deals, climbing sessions and chat between climbers.
- **Private by design:** Row Level Security in the database keeps private logs private.



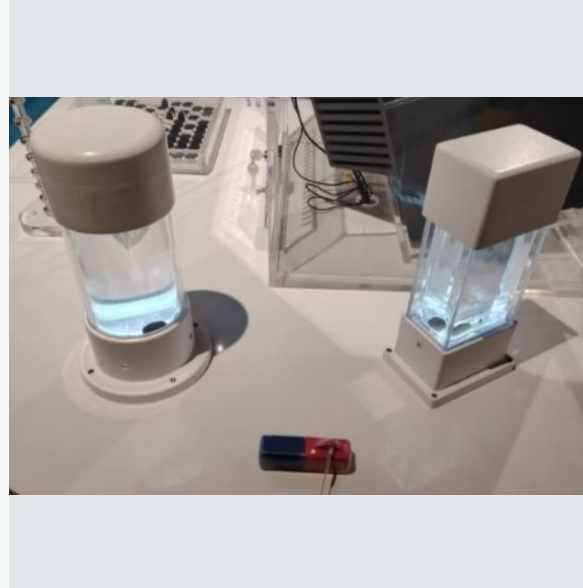
Overview



Modified microphone casing



Modified handheld UV light and holders



Redesigned casing with integrated light

MECHANICAL | WORK

SCS Some Call It Science

Setting: Science Centre Singapore

An installation of 11 interactive exhibits that shows visitors how science shapes everyday life.

I was involved in upgrading the exhibits: resolving issues and improving durability and reliability, including a modified microphone casing, a modified handheld UV light and holders, and a redesigned casing with an integrated light.



Visitor playing the game

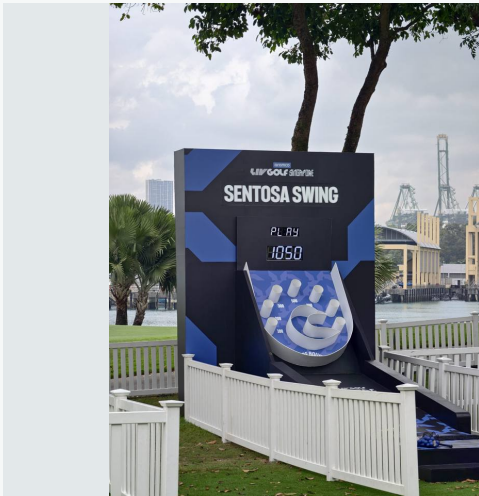
ELECTRONICS | WORK

LIV Golf Game Station

Setting: Sentosa, Singapore

Two interactive game booths, a golf game and a soccer game, for a public event held alongside the LIV Golf competition at Sentosa.

Both games are driven by IR sensors, built to be played by the public all day.



IR sensor-based golf game



IR sensor-based soccer game



Flip panel

MECHANICAL | WORK

Mandai Bird Park Installation

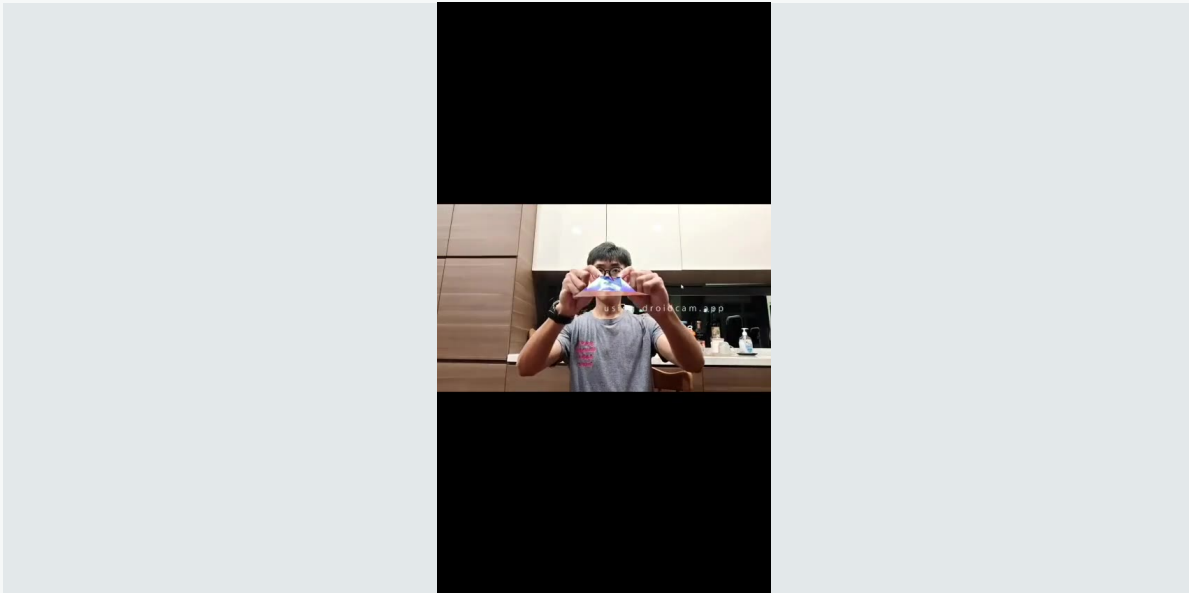
Setting: Mandai Bird Park

A flip information panel and an interactive mechanism that moves the flamingo's head to show how it feeds.

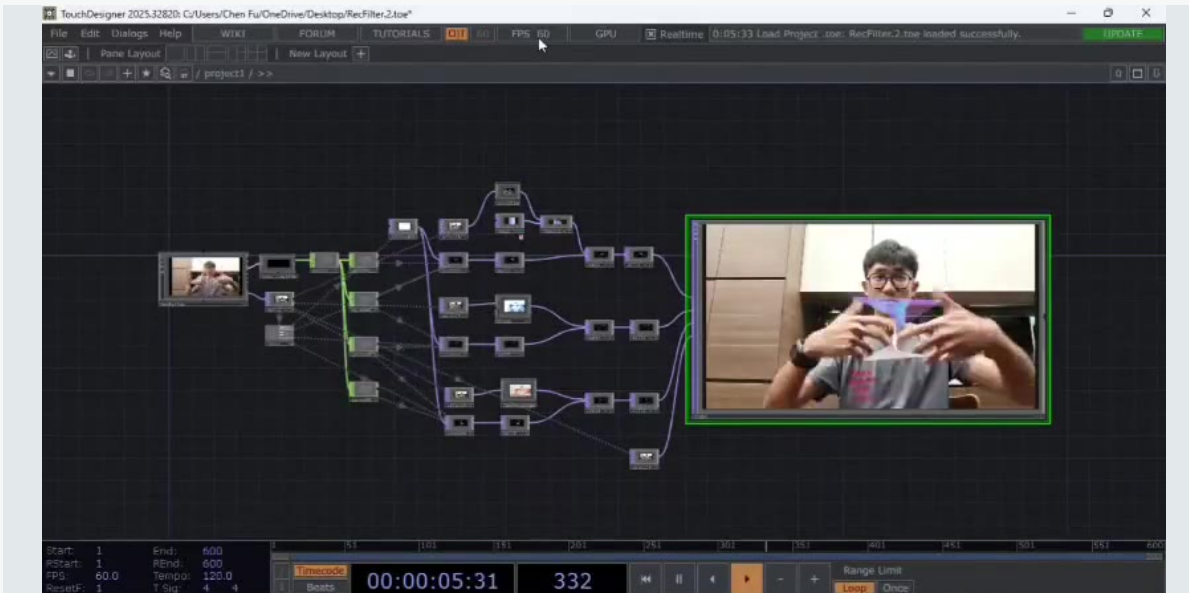
Visitors get an intuitive, hands-on way to learn about the bird.



Movable flamingo head



Visual output



Background system

OTHER | PERSONAL

Hand Gesture Visual Effect Control

Setting: MediaPipe, TouchDesigner

An AI camera system that recognises hand gestures and turns them into real-time visual effects for touch-free interaction.

The clips show the visual output and the background system running.

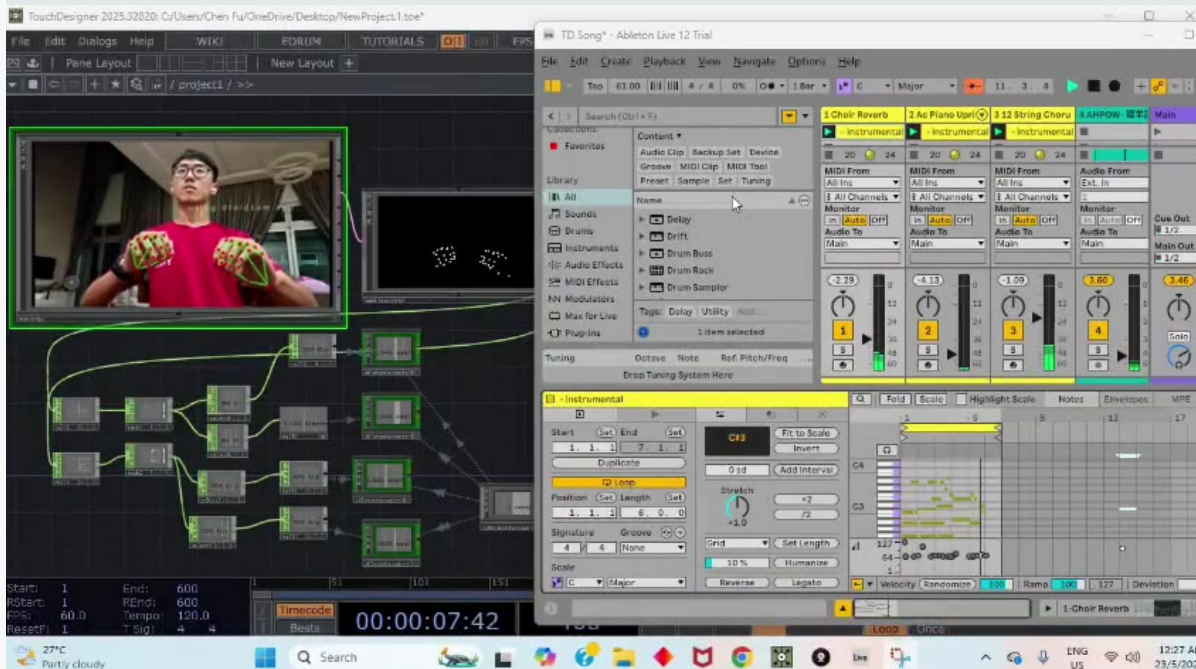
Video of this project is on the online portfolio.

OTHER | PERSONAL

Hand Gesture Audio Control

Setting: AI hand tracking, audio system

An interactive system that uses AI hand tracking through a camera so users can control different sound effects with hand gestures.



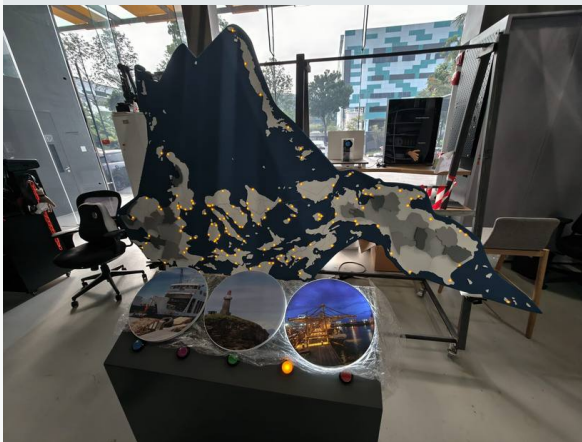
Video of this project is on the online portfolio.

LED Transport Map

Setting: Philippines

A transport map of the Philippines with five differently coloured LEDs, each controlled by its own button that also triggers a matching sound effect.

I completed the wiring and sound system setup, and supported the overall electronics design and integration of the interactive system.



Video of this project is on the online portfolio.



Visual animation

ELECTRONICS | WORK

NCCS Donor Wall Glowing Light

Setting: National Cancer Centre Singapore

A curved solid wall with many LED bulbs that create a glowing star around each donor's name, lighting the names with a warm, moving glow.

I designed and programmed the light animation.



Overview

Video of this project is on the online portfolio.

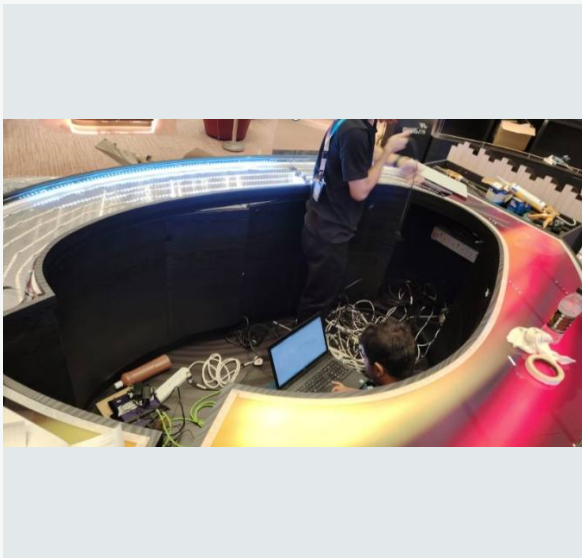
Bowmore Running Light Counter

Setting: Bowmore

A counter tabletop with an animated LED light box. Carefully arranged wiring and programmed LED sequences give a smooth, continuous running-light effect.



Final look



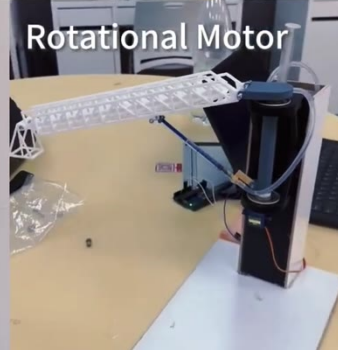
Extension Gear



Inclination Hydraulic



Rotational Motor



MECHANICAL | ACADEMIC

Telescopic Shore Gangway

Setting: INTI International University

A flexible gangway for LNG ship workers to embark, with three mechanical designs so its orientation and position can be adjusted for different shore set-ups.



Project poster

Video of this project is on the online portfolio.

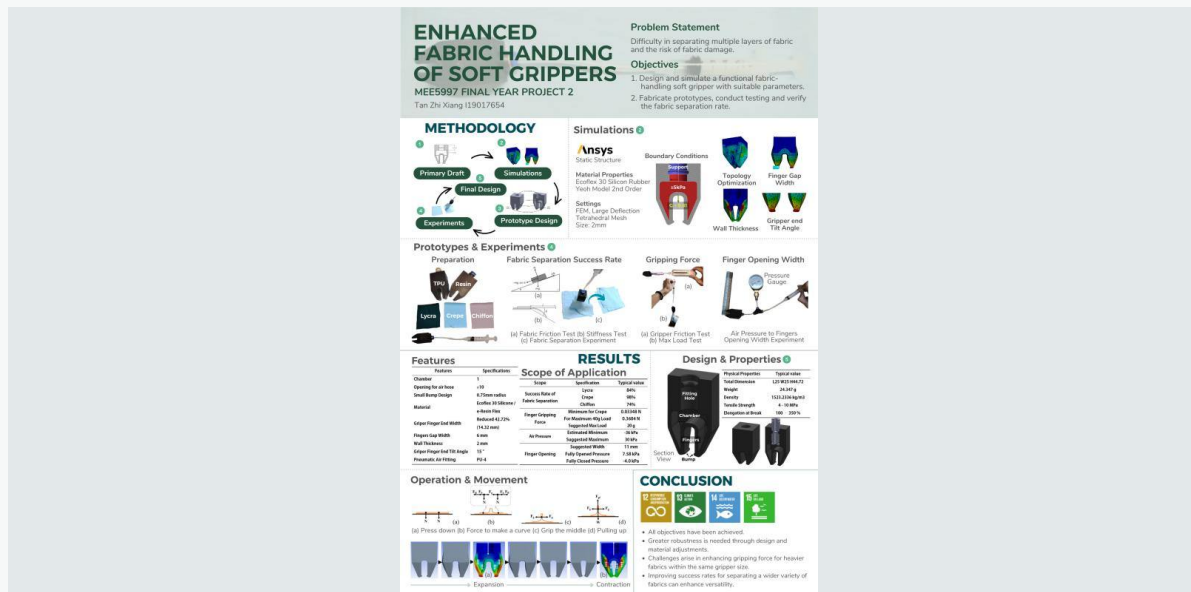


MECHANICAL | ACADEMIC

Pneumatic Soft Gripper

Setting: INTI International University

A soft pneumatic clamp made with 3D printing and a hollow-chamber design. It grips small, irregular objects, and separates fabric, without damaging their surfaces.



Project poster

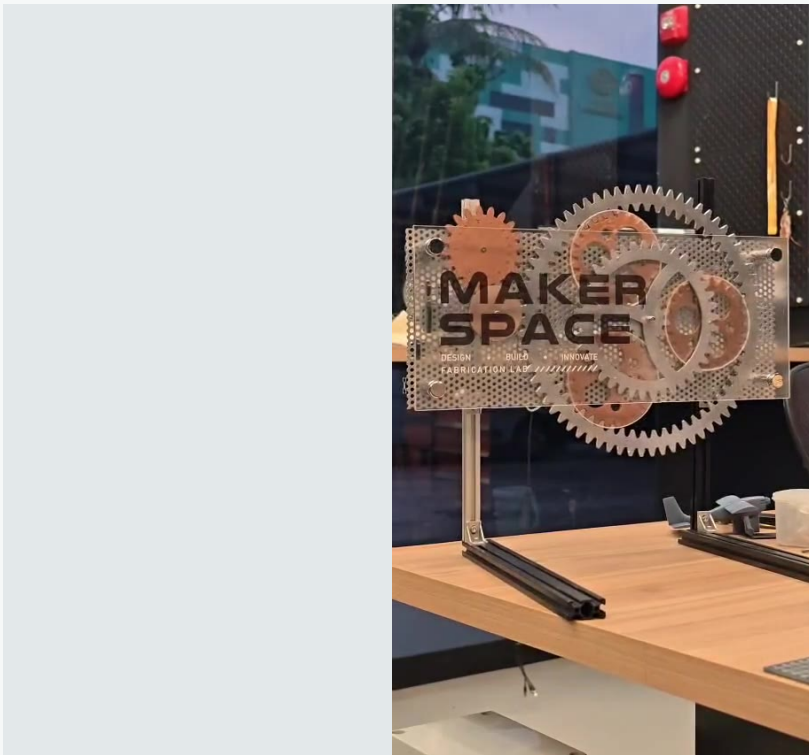
Video of this project is on the online portfolio.

MECHANICAL | WORK

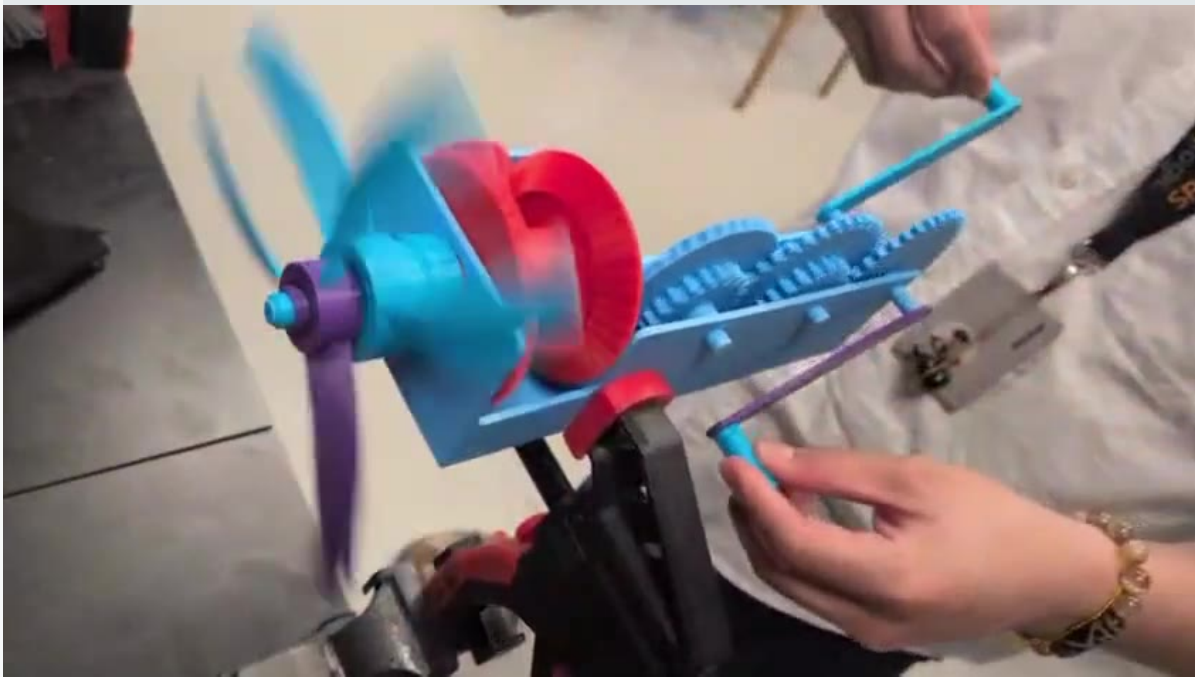
Planet Gear Signage

Setting: Maker space signage

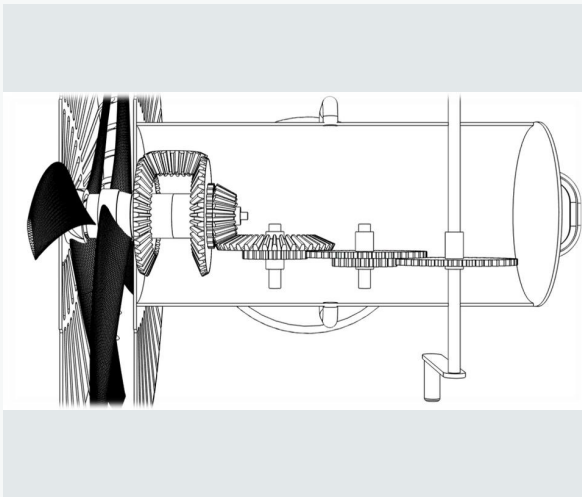
A metallic-look, 3D-printed planetary gear mechanism built as a moving sign that pairs mechanical motion with an eye-catching finish.



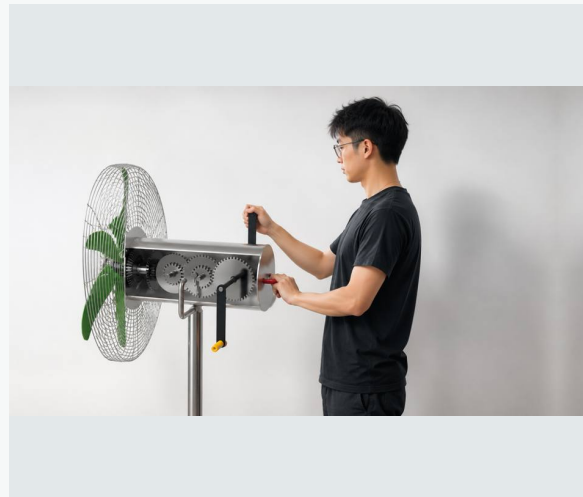
Video of this project is on the online portfolio.



Working prototype



Technical drawing



AI render

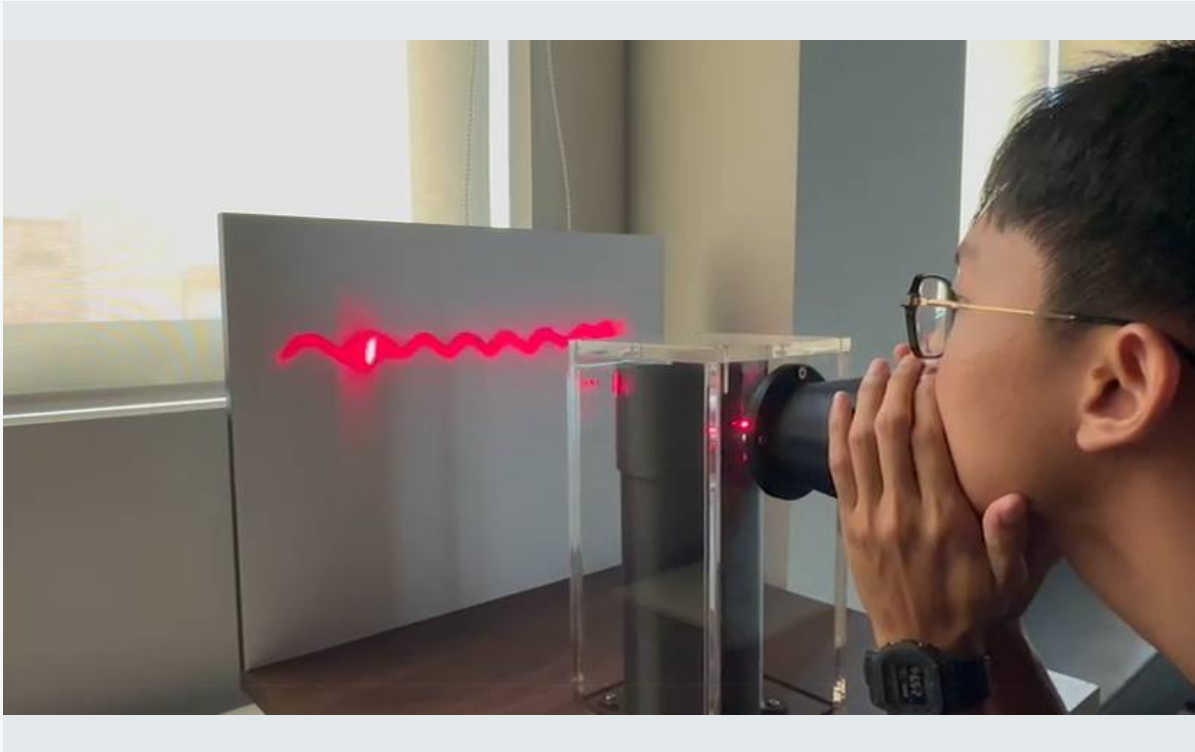
MECHANICAL | WORK

Handcrank Fan

Setting: Proof-of-concept mock-up

A mock-up that shows a speed step-up gear mechanism driving a counter-rotating blade, so you can see how the parts move and interact.

Video of this project is on the online portfolio.



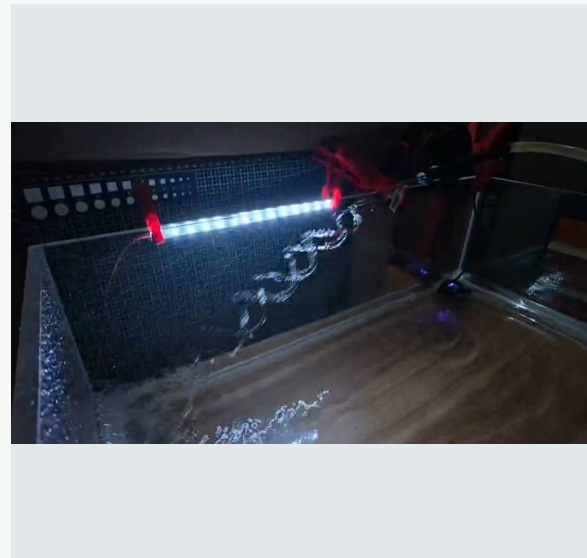
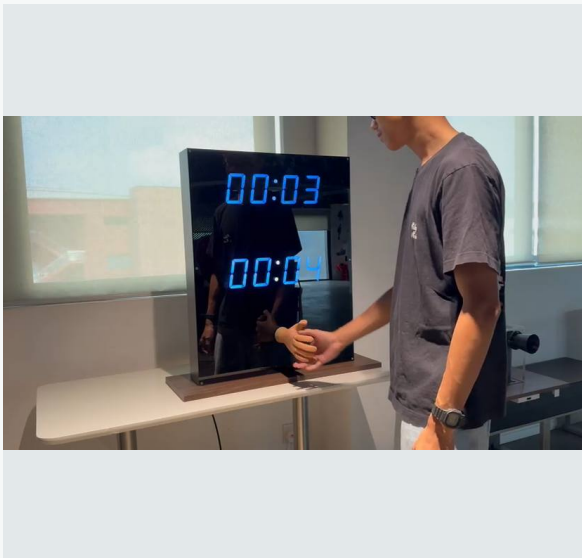
ELECTRONICS | WORK

SCS Fundamental Exhibits

Setting: Science Centre Singapore

Concept mock-ups built to demonstrate and validate the intended scientific effect and visitor experience before full builds.

They include a sound wave visualiser, a hand-holding experiment and a spiral water effect.

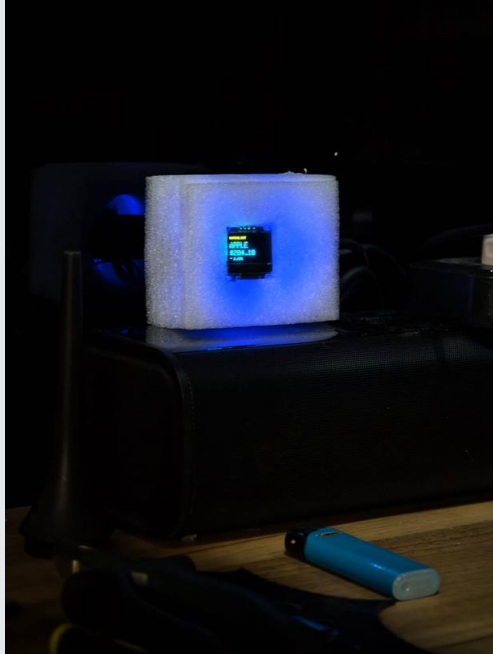
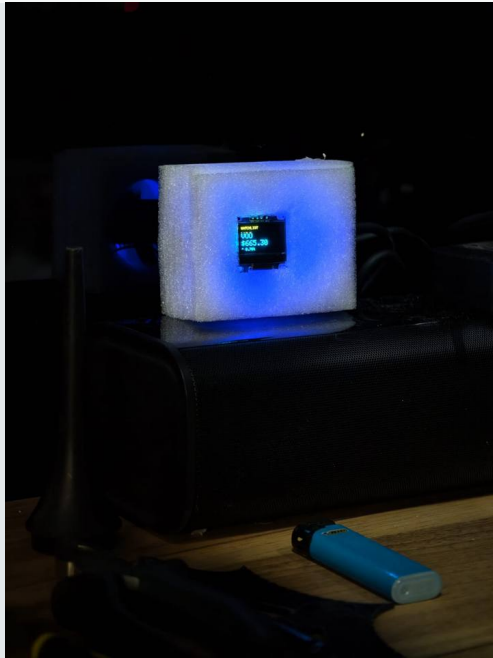


ELECTRONICS | PERSONAL

Real-time Stock Price Display

Setting: Wi-Fi, palm-sized

A Wi-Fi-connected, palm-sized device that pulls real-time stock market data and shows live prices on its built-in screen.

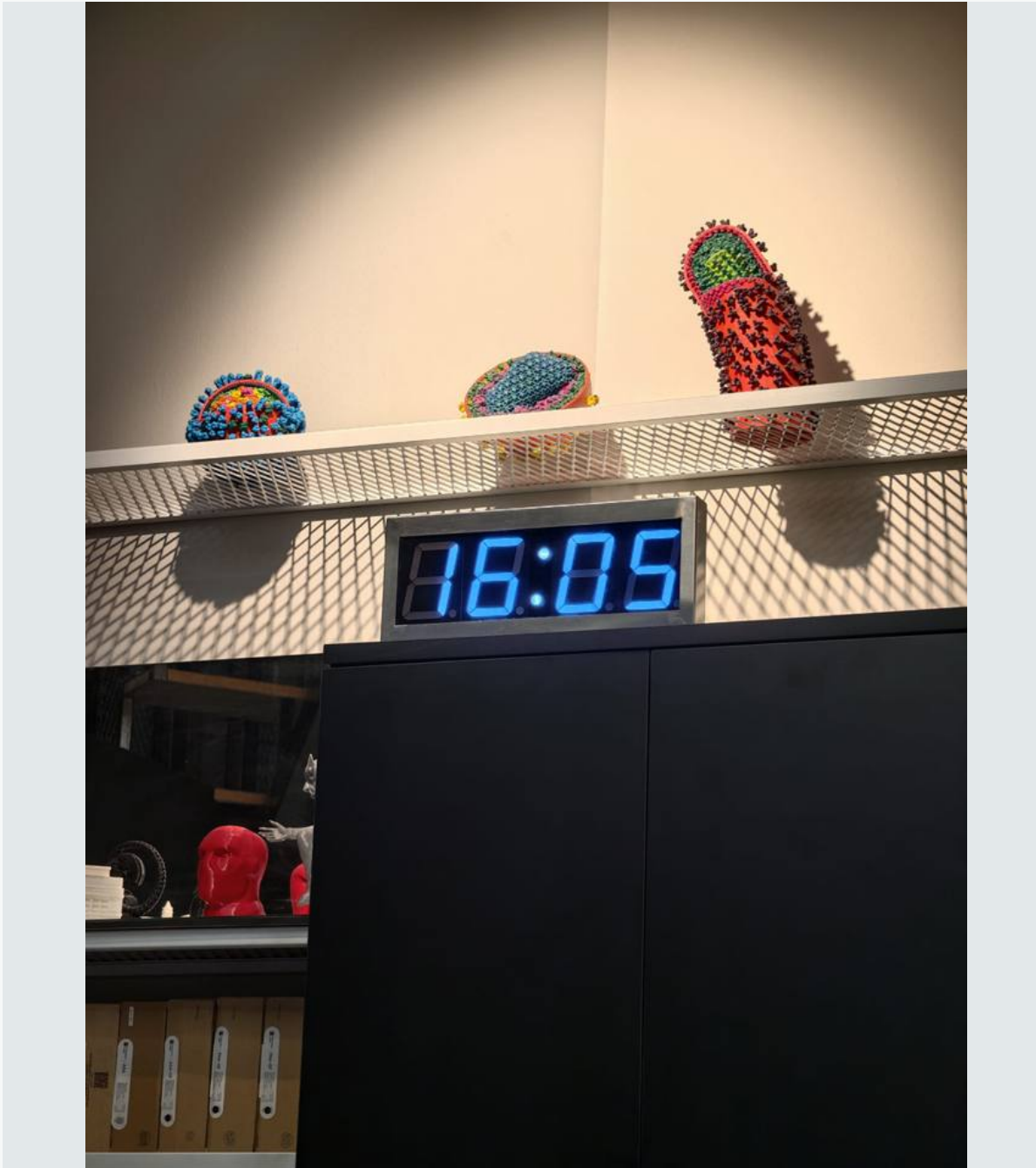


ELECTRONICS | PERSONAL

7-Segment Digital Clock

Setting: Bluetooth, smartphone sync

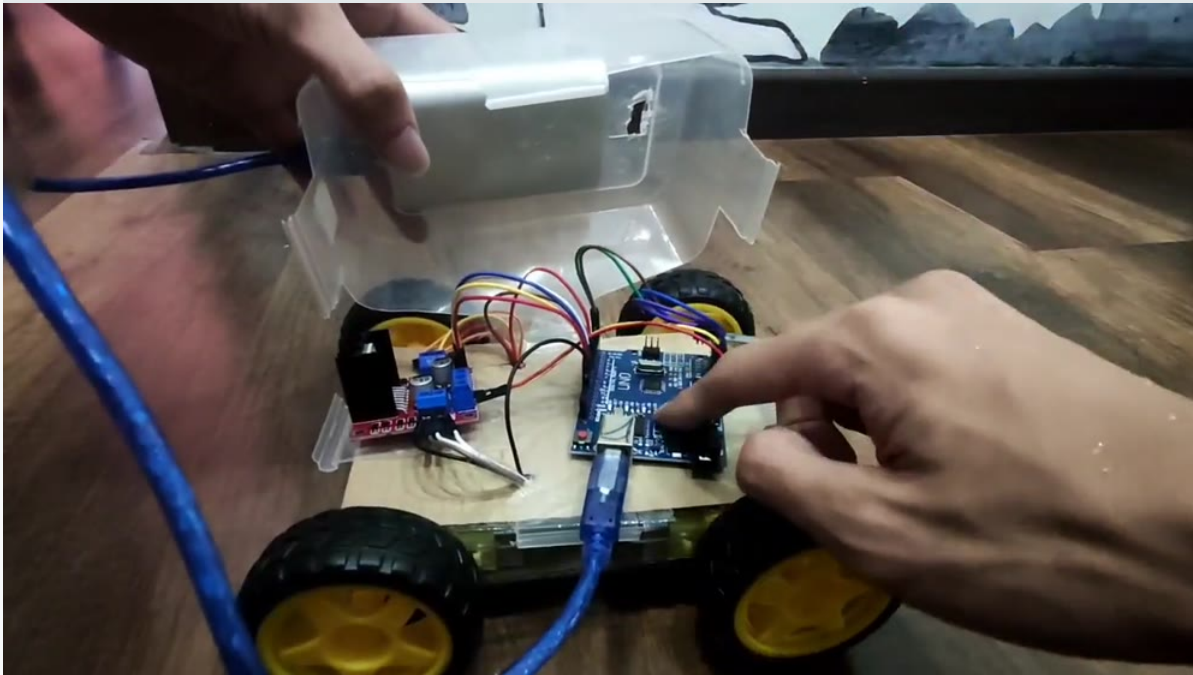
A smart digital clock that connects to a smartphone over Bluetooth, so the time syncs automatically and its settings can be changed from the phone.



Bluetooth-Controlled Car

Setting: INTI International University

A toy car operated from a smartphone, with wireless control of turning, acceleration and movement. The design is omnidirectional.



Video of this project is on the online portfolio.